

PROJECT PROPOSAL & SYNOPSIS

SDET Capstone Automation Project | Tricentis Demo Web Shop

1. Project Overview

Project Title	Enterprise E-Commerce Playwright Automation Framework
Target Application (SUT)	Demo Web Shop (Tricentis)
Application URL	https://demowebshop.tricentis.com
Technology Stack	Node.js, Playwright Test, Vanilla JavaScript, Dotenv, Allure Reports, GitHub Actions
Total Modules In Scope	8 Key Business Services
Total Test Cases	40 test cases across core workflows

2. Test Coverage & Module Breakdown

#	Module	Test Cases	Type Coverage
01	Authentication	15	Form Validation, Security, UI
02	Product Search	15	UI, Interaction, Data Validation
03	Product Details	15	UI, Arithmetic, Data Validation
04	Shopping Cart	15	Arithmetic, UI, Navigation
05	Wishlist Module	15	UI, Security, Navigation
06	Checkout	15	Form Validation, Navigation, E2E
07	Payment	15	Form Validation, Security, E2E
08	User Profile / Order History	15	UI, Form Validation, Navigation

3. Objective & Engineering Goals

The primary objective of this Capstone Project is to design, engineer, and deploy a highly resilient, scalable, and fast test automation framework from scratch using Playwright. The framework automates end-to-end user journeys on a real-world e-commerce portal, validating all business-critical modules against strict quality benchmarks.

- Zero-Flakiness Testing: Achieve stable execution by enforcing strict auto-waiting and web-first assertions.
- High Reusability - Page Object Model: Minimize code duplication by strictly separating web elements and business actions from test validation scripts using a clean POM architecture.
- Automated Cloud Quality via CI/CD: Integrate continuous quality gates that automatically trigger the full test suite in GitHub Actions.
- Data-Driven Robustness: Ensure complete test independence by utilizing dynamic test data generation to prevent duplicate record collisions.

4. Technology Stack Overview

Tool/Technology	Purpose & Framework Role
Playwright Test	Core browser automation engine providing native high-speed auto-waiting and web-first assertions.
Node.js / JavaScript	Native execution runtime and programming language; matches modern web applications.
Dotenv	Decouples and secures environment credentials, base URLs, and secrets in local env config files.
Allure Reports	Generates rich, graphic-based HTML test reports, trend dashboards, and suite-level analytics.
GitHub Actions	Remote CI/CD pipeline that executes the full test suite automatically in cloud containers.

5. Project Folder Structure

PATH	LAYER / PURPOSE
.github/workflows/playwright.yml	CI/CD Layer - GitHub Actions cloud execution pipeline workflow definition.
pages/	POM Layer - Contains all Page Object Model classes (e.g., LoginPage.js, CartPage.js).
tests/	Test Layer - Orchestration of test scenarios grouped by feature (e.g., auth.spec.js).
utils/	Utilities Layer - Dynamic data generators and shared helper functions.
playwright.config.js	Config - Master Playwright configuration: browsers, workers, timeouts, reporters.

MODULE 01: AUTHENTICATION

Objective: Validate user authentication functionalities (Registration, Login, Logout). | Total TCs: 15

TC ID	Test Title	Steps / Input	Expected Result	Type
TC_AUTH_01	Verify registration with valid details	Navigate to Register page; input valid First Name, Last Name, Email, Password, and Confirm. Click Register.	User is successfully registered and redirected to registration success page.	Form
TC_AUTH_02	Verify login with valid credentials	Navigate to Login page; input valid Email and Password. Click Log in.	User logs in successfully, email displays in header, and accesses dashboard.	Security
TC_AUTH_03	Verify login with invalid password	Navigate to Login page; input valid Email but incorrect Password. Click Log in.	System displays validation error: 'Login was unsuccessful. Please correct the errors and try again.'	Validation
TC_AUTH_04	Verify required field validation	Navigate to Login/Register page; leave mandatory fields blank and submit.	Form validation highlights empty fields with standard required field error messages.	Validation
TC_AUTH_05	Verify logout functionality	Log in with valid credentials; click the 'Log out' link in the top header.	Session terminates, user is redirected to homepage, and 'Log in' link reappears.	Security
TC_AUTH_06	Verify remember me functionality	Login using valid credentials and enable Remember Me checkbox	User session persists after browser restart	Security

TC_AUTH_07	Verify password minimum length validation	Enter short password during registration	Validation message appears for password length	Validation
TC_AUTH_08	Verify email format validation	Enter invalid email format during registration	System rejects invalid email format	Validation
TC_AUTH_09	Verify duplicate email registration	Register using already existing email	Registration blocked with duplicate email error	Security
TC_AUTH_10	Verify login using keyboard Enter key	Fill login form and press Enter key	Login process successfully executes	UI
TC_AUTH_11	Verify session expiration handling	Stay inactive after login for extended duration	Session expires securely after timeout	Security
TC_AUTH_12	Verify login page responsive rendering	Open login page on different viewport sizes	Login form renders correctly without UI breaking	UI
TC_AUTH_13	Verify multiple failed login attempts	Attempt invalid login multiple times	System continuously blocks invalid authentication	Security
TC_AUTH_14	Verify async login loader visibility	Trigger login request using valid credentials	Loading indicator appears until authentication completes	Async
TC_AUTH_15	Verify API login response status	Capture login API response during authentication	API returns successful status code and token	API

MODULE 02: PRODUCT SEARCH

Objective: Validate search functionality, valid/invalid keywords, and search results display.

| Total TCs: 15

TC ID	Test Title	Steps / Input	Expected Result	Type
TC_SRH_01	Search valid product keyword	Type 'laptop' or 'book' in global search bar and click Search.	Search results page loads displaying products matching the keyword.	UI
TC_SRH_02	Search invalid product keyword	Type 'invalid12345' in search bar and click Search.	System handles search gracefully without errors.	UI
TC_SRH_03	Verify no-result message	Submit search for non-existent item.	Displays message: 'No products were found that matched your criteria.'	Validation
TC_SRH_04	Verify search result count	Search for a generic term (e.g., 'computer'). Count displayed items.	Total product cards match the expected results count / pagination parameters.	UI
TC_SRH_05	Verify sorting functionality	On results page, select 'Price: Low to High' from the 'Sort by' dropdown.	Product cards dynamically reorder in ascending price format.	UI
TC_SRH_06	Verify search using partial keyword	Search using partial product name	Matching products display correctly	UI
TC_SRH_07	Verify search with uppercase characters	Search using uppercase keyword	Search remains case insensitive	Validation
TC_SRH_08	Verify special character search handling	Search using symbols or special characters	Application handles invalid symbols safely	Security
TC_SRH_09	Verify search suggestion dropdown	Type keyword slowly in search bar	Suggestions dynamically appear	Async

TC_SRH_10	Verify search result pagination	Search broad keyword with multiple pages	Pagination navigation works correctly	Navigation
TC_SRH_11	Verify search result product links	Open product from search results	User redirects to correct PDP	Navigation
TC_SRH_12	Verify sorting using price descending	Select High to Low sorting option	Products reorder correctly by price	UI
TC_SRH_13	Verify filtering after search	Apply filter after search execution	Filtered products update dynamically	UI
TC_SRH_14	Verify async product loading	Search product with slow network simulation	Loader displays until products fully load	Async
TC_SRH_15	Verify search API response structure	Capture search API payload	API returns valid product data schema	API

MODULE 03: PRODUCT DETAILS

Objective: Validate product-level information, visibility, and product interactions. | Total TCs: 15

TC ID	Test Title	Steps / Input	Expected Result	Type
TC_PDT_01	Verify product image is displayed	Navigate to a specific Product Detail Page (PDP).	Primary product image and thumbnails render fully without broken links.	UI
TC_PDT_02	Verify product price is correct	Open a PDP; extract and validate the price tag.	Price tag is visible, formatted correctly (e.g., decimals), and matches catalog view.	UI
TC_PDT_03	Verify quantity update functionality	Locate quantity input box on PDP; change value from 1 to 5.	Input accepts numeric values; updating quantity successfully registers for Add to Cart.	Form
TC_PDT_04	Verify product description visibility	Scroll down PDP to description and specification sections.	Full text details, manufacturer info, and specs render clearly.	UI
TC_PDT_05	Verify product availability	Check stock status tag on the PDP.	Tag accurately displays 'In stock' or 'Out of stock' preventing invalid purchases.	Validation
TC_SRH_06	Verify search using partial keyword	Search using partial product name	Matching products display correctly	UI
TC_SRH_07	Verify search with uppercase characters	Search using uppercase keyword	Search remains case insensitive	Validation
TC_SRH_08	Verify special character search handling	Search using symbols or special characters	Application handles invalid symbols safely	Security

TC_SRH_09	Verify search suggestion dropdown	Type keyword slowly in search bar	Suggestions dynamically appear	Async
TC_SRH_10	Verify search result pagination	Search broad keyword with multiple pages	Pagination navigation works correctly	Navigation
TC_SRH_11	Verify search result product links	Open product from search results	User redirects to correct PDP	Navigation
TC_SRH_12	Verify sorting using price descending	Select High to Low sorting option	Products reorder correctly by price	UI
TC_SRH_13	Verify filtering after search	Apply filter after search execution	Filtered products update dynamically	UI
TC_SRH_14	Verify async product loading	Search product with slow network simulation	Loader displays until products fully load	Async
TC_SRH_15	Verify search API response structure	Capture search API payload	API returns valid product data schema	API

MODULE 04: SHOPPING CART

Objective: Validate shopping cart quick actions, quantity updates, and subtotal arithmetic. |

Total TCs: 15

TC ID	Test Title	Steps / Input	Expected Result	Type
TC_CRT_01	Add product to cart	Click 'Add to cart' on a product from catalog or PDP.	Success notification displays; top header cart badge increments.	UI
TC_CRT_02	Remove product from cart	Open Shopping Cart; check 'Remove' checkbox for an item, click Update.	Item row disappears; cart totals recalculate immediately.	UI
TC_CRT_03	Verify cart quantity update	In cart review, change item QTY input box, click Update Cart.	Quantity modifies successfully without throwing errors.	Form
TC_CRT_04	Verify subtotal calculation	Add multiple items to cart with distinct prices; view subtotal.	Base price * quantity precisely matches the calculated Subtotal displayed.	Arithmetic
TC_CRT_05	Verify empty cart message	Remove all items from the cart.	Displays clean placeholder text: 'Your Shopping Cart is empty!'	UI
TC_CRT_06	Verify multiple product addition	Add multiple products into cart	Cart displays all added items	UI
TC_CRT_07	Verify cart persistence after refresh	Refresh cart page after adding items	Cart items remain saved	Security
TC_CRT_08	Verify cart badge count update	Add and remove products repeatedly	Badge updates dynamically	UI
TC_CRT_09	Verify subtotal after quantity change	Update item quantity multiple times	Subtotal recalculates accurately	Arithmetic

TC_CRT_10	Verify invalid quantity handling	Enter alphabetic quantity value	Invalid input gets rejected	Validation
TC_CRT_11	Verify remove all items functionality	Remove all cart items together	Cart becomes empty successfully	UI
TC_CRT_12	Verify navigation to checkout	Click Checkout button from cart	User redirects to checkout page	Navigation
TC_CRT_13	Verify cart calculation using reduce logic	Add products with varying quantities	Final subtotal calculation remains accurate	Arithmetic
TC_CRT_14	Verify async cart update loader	Update cart quantities rapidly	Loader appears during updates	Async
TC_CRT_15	Verify cart API synchronization	Monitor cart API requests	API updates cart successfully	API

MODULE 05: WISHLIST MODULE

Objective: Validate wishlist capabilities, saving items, and session persistence. | Total TCs:

15

TC ID	Test Title	Steps / Input	Expected Result	Type
TC_WSH_01	Add product to wishlist	Click 'Add to wishlist' button on a product card.	Success banner shows; wishlist header badge increments by 1.	UI
TC_WSH_02	Remove product from wishlist	Navigate to Wishlist page; check 'Remove' box for an item, update.	Item is successfully removed from the saved tabular list.	UI
TC_WSH_03	Verify wishlist item count	Add 3 distinct items to the wishlist.	Wishlist header badge and table row count accurately reflects '3'.	Validation
TC_WSH_04	Verify wishlist persistence	Add items to wishlist, log out, then log back into the same account.	Saved items remain securely in the wishlist across authenticated sessions.	Security
TC_WSH_05	Verify wishlist functionality after login	Add item as guest; log in to an existing account.	Guest wishlist items merge or transfer to the authenticated account's list.	Navigation
TC_WSH_06	Verify duplicate wishlist prevention	Add same product repeatedly	Duplicate entries are prevented	Validation
TC_WSH_07	Verify wishlist page refresh persistence	Refresh wishlist page	Saved products remain visible	Security
TC_WSH_08	Verify wishlist item navigation	Open product from wishlist	Correct PDP opens successfully	Navigation
TC_WSH_09	Verify wishlist to cart transfer	Move item from wishlist to cart	Product transfers successfully	UI
TC_WSH_10	Verify wishlist count after removal	Remove product from wishlist	Wishlist badge decrements correctly	Validation
TC_WSH_11	Verify multiple wishlist additions	Add several products to wishlist	All products display correctly	UI

TC_WSH_12	Verify async wishlist updates	Add products rapidly to wishlist	Loader handles async operations correctly	Async
TC_WSH_13	Verify wishlist table rendering	Open wishlist with multiple items	Table structure renders properly	UI
TC_WSH_14	Verify unauthorized wishlist access	Access wishlist URL without login	User redirects to login page	Security
TC_WSH_15	Verify wishlist API response	Capture wishlist API request	API returns correct wishlist data	API

MODULE 06: CHECKOUT

Objective: Validate the complete multi-step checkout workflow and address mapping. |

Total TCs: 15

TC ID	Test Title	Steps / Input	Expected Result	Type
TC_CHK_01	Verify checkout page loads	From cart, click 'Checkout' button (while authenticated).	Navigates to secure checkout stepper starting at Billing Address step.	Navigation
TC_CHK_02	Verify shipping method selection	Proceed to Shipping Method step; select Ground vs Next Day Air.	Radio button registers selection; cost updates bind to final order total.	Form
TC_CHK_03	Verify billing address validation	Attempt to proceed from Billing Address step with missing required fields.	Step locks; validation errors highlight missing mandatory dropdowns/text fields.	Validation
TC_CHK_04	Verify order summary details	Reach the 'Confirm Order' step; review cart matrix.	Itemized products, prices, shipping, and tax match the original cart invoice.	Arithmetic
TC_CHK_05	Verify checkout navigation	Click 'Back' or previous stepper headers during checkout flow.	System allows safe backward navigation without losing previously entered data.	Navigation
TC_CHK_06	Verify checkout without login	Click Checkout as guest user	System redirects user to Login page	Security
TC_CHK_07	Verify shipping address update	Modify shipping address during checkout	Updated address saves successfully	Form
TC_CHK_08	Verify invalid ZIP code validation	Enter invalid ZIP/postal code	Validation error message appears	Validation
TC_CHK_09	Verify country dropdown selection	Select different countries in billing section	State and region fields update properly	UI

TC_CHK_10	Verify checkout progress stepper	Navigate through all checkout steps	Stepper highlights current active step	Navigation
TC_CHK_11	Verify tax calculation accuracy	Add products with taxable values	Tax calculation matches expected total	Arithmetic
TC_CHK_12	Verify shipping charge updates	Change shipping methods repeatedly	Final order total updates dynamically	Arithmetic
TC_CHK_13	Verify checkout form field persistence	Navigate backward during checkout	Previously entered form data remains	Navigation
TC_CHK_14	Verify async checkout processing loader	Proceed between checkout steps rapidly	Loader appears while processing requests	Async
TC_CHK_15	Verify checkout API response	Capture checkout API payload	API successfully returns order summary details	API

MODULE 07: PAYMENT

Objective: Validate payment method selection, logic paths, and final transaction success. |

Total TCs: 15

TC ID	Test Title	Steps / Input	Expected Result	Type
TC_PAY_01	Verify payment page loads	Proceed from Shipping Method to Payment Method step.	Payment options (e.g., COD, Check/Money Order, Credit Card) render properly.	UI
TC_PAY_02	Verify payment method selection	Select a specific payment method (e.g., Cash On Delivery) and click Continue.	System registers method and advances to Payment Information step.	Form
TC_PAY_03	Verify card number validation	Select Credit Card method; input invalid alphanumeric card strings.	Client-side validation rejects invalid formats prior to submission.	Validation
TC_PAY_04	Verify payment confirmation message	Complete all steps and click 'Confirm'.	Navigates to success view: 'Your order has been successfully processed!'	Navigation
TC_PAY_05	Verify invalid payment details handling	Provide expired dates or invalid CVV on mock credit card form.	Form throws validation block; prevents transition to final confirmation step.	Security
TC_PAY_06	Verify multiple payment methods visibility	Open Payment Method section	All supported payment methods display correctly	UI
TC_PAY_07	Verify expired card validation	Enter expired credit card details	System blocks invalid card submission	Validation
TC_PAY_08	Verify CVV length validation	Enter invalid CVV length	Validation error displays correctly	Validation

TC_PAY_09	Verify cardholder name mandatory validation	Leave cardholder name blank	Payment submission gets blocked	Validation
TC_PAY_10	Verify payment cancellation flow	Cancel payment before confirmation	User safely returns to previous step	Navigation
TC_PAY_11	Verify payment retry functionality	Submit invalid payment then retry valid one	Payment successfully processes after retry	Security
TC_PAY_12	Verify order total during payment	Compare cart total with payment summary	Payment total matches checkout summary	Arithmetic
TC_PAY_13	Verify async payment processing loader	Confirm order using valid payment details	Processing loader appears until completion	Async
TC_PAY_14	Verify secure payment field masking	Enter card details in payment form	Sensitive fields remain masked securely	Security
TC_PAY_15	Verify payment API transaction response	Capture payment API response	API returns successful transaction status	API

MODULE 08: USER PROFILE / ORDER HISTORY

Objective: Validate account detail tracking, history logs, and session consistency. | Total TCs: 15

TC ID	Test Title	Steps / Input	Expected Result	Type
TC_USR_01	Verify user profile details	Navigate to 'My Account' -> Customer Info.	First Name, Last Name, and Email accurately match registration details.	UI
TC_USR_02	Verify previous orders are displayed	Complete a purchase, then navigate to 'Orders' tab in My Account.	Newly generated order is listed with correct Date, Total, and Order Status.	UI
TC_USR_03	Verify account information update	Change Last Name in Customer Info and click Save.	Data updates successfully; changes persist upon page refresh.	Form
TC_USR_04	Verify order history page rendering	Click 'Details' on a past transaction listed in the Orders view.	Detailed invoice page loads showing precise billing, shipping, and product breakdowns.	Navigation
TC_USR_05	Verify user session persistence	Log in, close browser tab (not logout), reopen browser and navigate to URL.	User session remains active securely based on cookie/token limits.	Security
TC_USR_06	Verify profile page accessibility	Open My Account page after login	User profile page loads successfully	UI
TC_USR_07	Verify email field readonly behavior	Attempt editing registered email field	Email field remains protected/read-only	Security
TC_USR_08	Verify password change functionality	Update account password using valid inputs	Password updates successfully	Security

TC_USR_09	Verify invalid profile update handling	Submit profile form with empty mandatory fields	Validation messages appear correctly	Validation
TC_USR_10	Verify order detail navigation	Open details of previous order	Correct invoice details page loads	Navigation
TC_USR_11	Verify order sorting functionality	View multiple order history entries	Orders display in latest-to-oldest order	UI
TC_USR_12	Verify profile data persistence after refresh	Update profile and refresh page	Updated information persists correctly	Security
TC_USR_13	Verify async order history loading	Open order history with multiple records	Loader appears until records fully render	Async
TC_USR_14	Verify logout from profile section	Logout using My Account page	User session terminates successfully	Security
TC_USR_15	Verify profile API response structure	Capture customer profile API response	API returns valid customer information schema	API